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Enhancing Drug Recommendations: A Modified LSTM Approach in Intelligent Deep Learning Systems

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Abstract

The overwhelming burden on the healthcare system has resulted in increased mortality rates as individuals struggle to receive appropriate medical attention. Machine learning has demonstrated its efficacy across diverse applications, fueling a growing trend in the realm of innovative automation. This is particularly facilitated by drug recommendation systems, which possess the capability to significantly reduce the dependence on specialized assistance. Faced with limited access to medical expertise, a considerable number of individuals have turned to self-medication, often worsening their health conditions. In response to this critical scenario, this paper introduces an inventive drug recommendation system which helps to lessen the strain healthcare professionals. Here, we present the development of a medication recommendation system based on LSTM, utilizing user-provided symptoms and the model's evaluations of past drug recommendations. The results of the comparison between Random Forest, SVM, and LSTM indicated that the modified LSTM-based recommendation system outperforms the other two models, achieving a 94% accuracy rate.

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Keywords: Drug Recommendation; Drug Classification; Long Short-Term Memory (LSTM); Support Vector Machine (SVM); Random Forest Classifier.

1. Introduction

In the endeavor to enhance healthcare accessibility and address the evolving challenges of obtaining appropriate medical attention, this research aims to introduce an innovative and intelligent drug recommendation system. The

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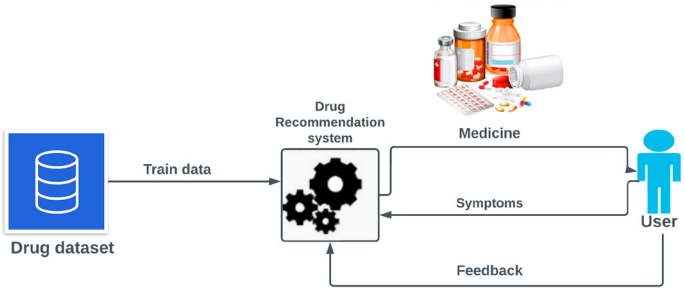


Fig. 1. Drug Recommendation System

escalating burden on the healthcare system necessitates creative solutions to reduce dependence on specialized medical assistance.

The majority of approaches for drug recommendation rely on content-based or collaborative filtering models. Collaborative filtering makes use of user interactions, suggesting drugs based on the preferences of users with similar tastes or item similarities. On the other hand, content-based filtering examines drug features and user preferences to offer recommendation

We design the system to assist individuals who may have limited access to medical expertise and resort to self-medication, potentially exacerbating their health conditions. The target users are those seeking a reliable and intelligent drug recommendation system.

The dataset is meticulously crafted by extracting comprehensive information from the authoritative CIMS Drug Monitor book. It encompasses vital details such as indications, dosage, contraindications, and adverse reactionsThe Extracted data is raw and has various unecessary words . The textual data undergoes essential preprocessing techniques, including tokenization and the application of bag-of-words approaches.A key method in Natural Language Processing (NLP)[1] [2] is tokenization, which divides textual input into smaller pieces known as tokens. Tokenization is used in our context to extract meaningful units for further analysis from each drug's indications. This procedure makes it easier to transform unstructured text into a format that machine learning algorithms can understand.Sentiment Analysis[3] is used to analyze the tokenized words . We can improve a recommendation framework by gaining insights into the user experience by measuring the sentiment associated with medication reviews.

Machine learning approaches, such as regression, classification, and clustering, are widely used in the healthcare domain. Each serves a specific purpose in solving diverse problems. Regression, for instance, predicts continuous numeric values, while classification categorizes observations into groups or classes.

The classification algorithms[4] predict the category of new observations based on training data. Support Vector Machines, Artificial Neural Networks, Logistic Regression, Random Forest, and Naïve Bayes are popular classification algorithms in the healthcare domain.SVM[5] is a supervised machine learning algorithm used for both regression and classification problems. It finds the optimal hyperplane in an N-dimensional space to classify provided data points. Random Forest builds decision trees on various random samples of the dataset, taking a majority vote for classification.

Random Forest[6] is one of the machine learning algorithms which is widely used for regression and classification problems. The main intuition behind random forest algorithm is that it builds decision trees on various different random samples of the dataset and takes the majority vote for classification problems and mean or average in the case of regression problems. LSTM, stands out as a distinctive type of Recurrent Neural Network (RNN) designed to grasp long-term data dependencies. Its ability to achieve this stems from the intricate architecture of the recurrent model module, comprising four layers that interact collaboratively.

Despite significant progress in recent years, technological ML models still suffer from a number of inefficiencies that can have a significant impact on their performance and reliability in real-world healthcare applications.These inefficiencies can limit the potential of ML models to improve healthcare delivery and outcomes.

The importance of this research lies in addressing the overwhelming burden on the healthcare system. The primary goal is to present an inventive drug recommendation system based on a modified LSTM approach. The system utilizes user-provided symptoms and the model's evaluations of past drug recommendations to offer accurate and efficient

medication suggestions. Fig 1 depicts the overview of the proposed Drug Recommendation System. In natural language processing, LSTMs have been used to improve the performance of language models, machine translation, and text summarization [7]. The modified LSTM comprises a four-layered architecture, fostering intricate interactions to enhance overall system performance. This refined LSTM model is deliberately selected to optimize the efficiency and effectiveness of the drug recommendation system.In evaluating our model, matrices play a crucial role, offering detailed insights into its performance. The confusion matrix breaks down True Positives, True Negatives, False Positives, and False Negatives, providing a nuanced understanding [8]. Personally, we find the F1 score to be a valuable metric, as it harmonizes precision and recall, offering a comprehensive assessment of our model's efficacy. Additionally, accuracy, representing the proportion of accurately predicted instances to the total instances, stands as a fundamental metric for gauging the overall effectiveness of our model.The system is designed to assist individuals who may have limited access to medical expertise and resort to self-medication, potentially exacerbating their health conditions. The target users are those seeking a reliable and intelligent drug recommendation system.The modified LSTM, a specialized variant of recurrent neural networks, allows our model to capture and understand long-term dependencies within the data, enhancing its ability to provide more accurate and context-aware drug recommendations.

Our primary contribution of this work is as follows

- Introducing a creative drug recommendation model with a modified LSTM architecture.
- Employing sentiment analysis to enhance the efficiency of the model.
- Establishing a recommendation system supported by feedback to address the dataset shift problem.

The paper is structured as follows:Section 2 delves into the literature survey, exploring existing research in the field, while Section 3 outlines the methodology, providing a detailed account of the research approach. In Section 4, the results are comprehensively discussed, shedding light on the outcomes and their interpretations. Even though we cannot completely remove Doctors from the health care. The suggested deep LSTM RNN architecture outperforms conventional LSTM networks and DNNs, effectively utilizing model parameters to enhance performance. This improvement is particularly notable in addressing the computational efficiency needed for training extensive networks.

2. Literature Survey

In this subsection, we conduct a comprehensive review of prior research within related domains. This includes an exploration of various machine learning algorithms employed for text classification, encompassing not only conventional methods but also specialized LSTM models. Additionally, we scrutinize the evaluation techniques applied in these studies, emphasizing a formal examination of the literature in alignment with the specified research focus.

Text data is one of the most popular examples of sequential data[9] . Before we can input our input text data into a machine learning algorithm, we may need to perform a variety of preprocessing steps depending on the nature of the data. Text classification, also called text categorization, is a traditional technique in NLP, which aims to assign classes or labels to text data such as paragraphs, sentences, or documents. There are two approaches in automatic text classification which are rule-based methods and machine learning-based methods.

One of the most performed operation is tokenization.[10] Tokenization is the process of breaking down our text data into smaller sections. There involves two types of tokenization which are sentence tokenization and word tokenization [11]. Sentence tokenization involves breaking down the text data into individual sentences whereas Word tokenization[12] involves breaking down the text data into individual words. We can either use the split function in python to perform this operation or we can use the tokenize function in NLTK library.Another import preprocessing step is the removal of unwanted special symbols or characters. We can use regular expression to perform this operation.Hutto et al,[13] In the particular paper which discuss about the use of vader in twitter sentiment analysis[14].The paper opens the opportunities that vader can provide .This methods explores a suggested system tool designed for Twitter data, aiming to automatically analyze Twitter posts and categorize them as positive, negative, or neutral sentences. To accomplish this task, the authors employ techniques like natural language processing, pattern recognition, and predictive methods.

Choosing which specific machine learning algorithm we should use is one of the most critical task. The classification algorithms evaluation is most often relied on accuracy of the predicted values. There are at least three techniques which are widely used in machine learning tasks to calculate the classifier's accuracy.

One employed technique involves partitioning the training set, allocating two-thirds for training and reserving the remaining third for performance evaluation. In another approach, known as cross-validation, the dataset is segmented into mutually exclusive and equally sized subsets. For each subset, the classification algorithm is trained on the combined remaining subsets, and the mean error rate of each subset serves as an estimate for the model's error rate. we comprise some of the works that have been done in the field of supervised machine learning techniques[15] . There are numerous number of applications of Machine Learning [16] , the most prominent of which is data mining. Inductive machine learning is a technique of learning a number of rules from given instances (data in the training set), or we could say that, making a classification model that can be used in the application of generalizing from new instances. A variety of factors must be considered and examined carefully: perhaps the most relevant features in the dataset are not being used or a larger dataset is needed or the dimensionality is too big or the selected machine learning algorithm is inappropriate or hyper parameter tuning may be required.

When compared to feed forward neural networks, LSTM [17][18] has feedback connections therefore it has the ability to learn long-term dependencies. However, these networks encounter a challenge related to parameter tuning. Normally, the initial and control parameters of the LSTM network are selected on a trial and error basis. In recent developments, scholars have successfully devised methods for reducing label space dimensions to categorize text with multiple labels. Nonetheless, a considerable number of these designed models tend to overlook the intricacies within the sequential aspects of texts and the correlation between the labels and the original label space. Thus, it was assumed that these labels were just vectors with no significance [19]. LSTMs are designed to control the flow of gradients during the training process, allowing for more stable and efficient learning. This is crucial in tasks where the prediction or recommendation depends on a sequence of symptoms or events.However these models often demand significant computational power. Recently, there has been a rise in the development of multi-label text classification models, which are notably more intricate compared to their single-label counterparts. Presently, 03 researchers widely employ deep learning models like RNN and LSTM[20] for text classification. However, these models encounter challenges, particularly when dealing with imbalanced datasets.The significance of LSTM in sequential data processing, particularly in drug recommendation systems, lies in its ability to address the vanishing and exploding gradient problems associated with traditional RNNs. These problems hinder the effective learning of long-range dependencies in sequences

In healthcare, traditional machine learning techniques are commonly applied to precision medicine, aiming to predict the treatment protocols that are more likely to succeed for a patient based on various patient attributes and the contextual details of the treatment.[21]. Most machine learning and precision medicine applications heavily depend on having a labeled training dataset where the outcome or target variable is already known, constituting the supervised learning approach.

A Deep Neural Network is another variant of ML techniques [22] that has been introduced in the late 1960s, deep neural networks have found applications in categorization, such as predicting a patient's susceptibility to a specific disease. These networks excel in framing problems by considering inputs, outputs, and variable weights or features that connect inputs to outputs. While there's a loose analogy to how neurons process signals, it's essential to note that the comparison to the brain's function is relatively weak Neural Networks or deep learning[23] is also increasingly being used in speech recognition and other applications of Natural Language processing. Unlike the other forms of statistical analysis, all features in a deep learning model usually has little to no meaning when observed by a human. As such, the explanation of the model's prediction may be very difficult or near to impossible to interpret by a human.

There are various metrics for evaluating classification models in machine learning [20] . One of the most popular approach is to check the accuracy. How often the classifier predicts outputs accurately may be determined by looking at accuracy. There are advantages and disadvantages for accuracy. The main disadvantage of evaluating the model with accuracy is that it neglects the differences between the types of errors and another disadvantage is the dependency on the distribution of class in the dataset.Confusion matrix is another one of the most popular Generally we use confusion matrix To summarize the performance of a particular classification model. It shows the ways in which your classification algorithm is confused when it makes prediction. It gives you the insight not only into the errors being made by your classification model but most importantly the types of errors that are being made.Another model

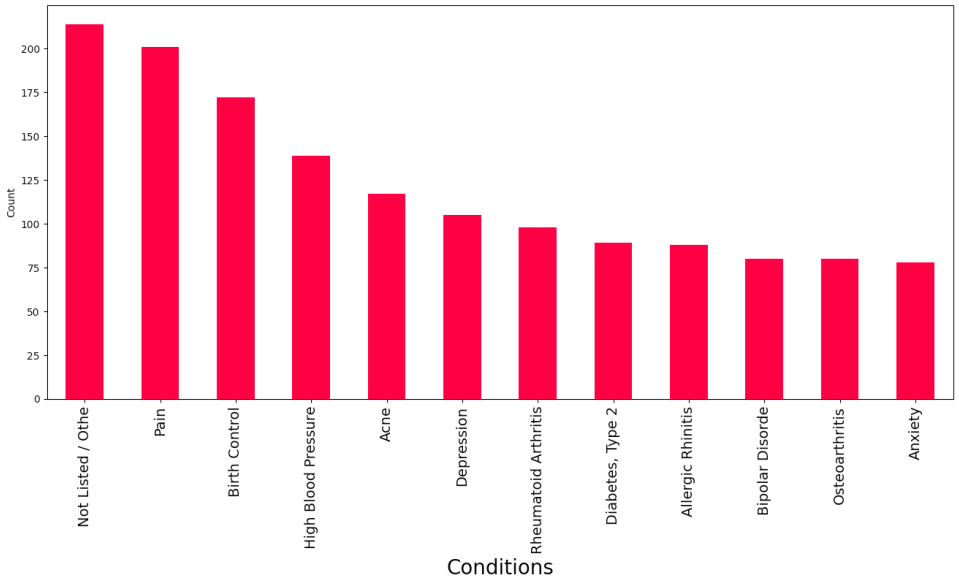


Fig. 2. Bar plot depicting the top 12 medical conditions with the highest number of available drugs.

evaluation method is to check Precision. Precision is basically how close or dispersed the measurements made are to each other. Its value presents the fraction of predictive positive values that are accurate.

This review explores text classification, covering machine learning algorithms, including LSTM models. It emphasizes preprocessing importance and discusses challenges in algorithm selection, supervised learning, and LSTM's handling of long-term dependencies. In healthcare, traditional machine learning aids precision medicine, and deep neural networks pose interpretability challenges. Evaluation metrics like accuracy and confusion matrix are highlighted, acknowledging their strengths and limitations. The review stresses nuanced considerations in text classification, from algorithm choices to model evaluation challenges.

3. Proposed Methodology

In this section, the comprehensive workflow of the drug recommendation system is explained and illustrated in Fig5 . It encompasses three primary stages: Data Collection and Processing, Model Generation, and Testing. The initial phase involves data collection and processing.It comprises two distinct steps: drug data retrieval and data processing. These two phases are intricately detailed to provide a comprehensive understanding of the system's functioning. In Data Collection phase, the drug information is collected from various sources and stored.The selection of the CIMS Drug Monitor book as our substrate is a strategic choice deeply impacting the richness of our dataset. This comprehensive medical reference source encapsulates a wealth of information crucial for drug recommendation systems, including indications, dosage, contraindications, and adverse reactions. The dataset's fidelity to authoritative medical knowledge enhances the model's ability to discern nuanced patterns and relationships. This substrate choice aligns with our goal of capturing relevant features for more accurate predictions in drug recommendation. In Preprocessing phase phase, we build a classification model using data mining techniques that are able to classify the drugs based on the drug data collected from our earlier data retrieval phase.The detiailed overview is provided in the coming section

3.1. Data Collection and Processing

The strategic choice of the CIMS Drug Monitor book as the substrate for dataset creation significantly influences the richness and quality of our dataset. This comprehensive medical reference source serves as a goldmine of information crucial for the development of robust drug recommendation systems. It provides detailed insights into indications, dosage, contraindications, and adverse reactions, offering a holistic view of pharmaceutical knowledge.

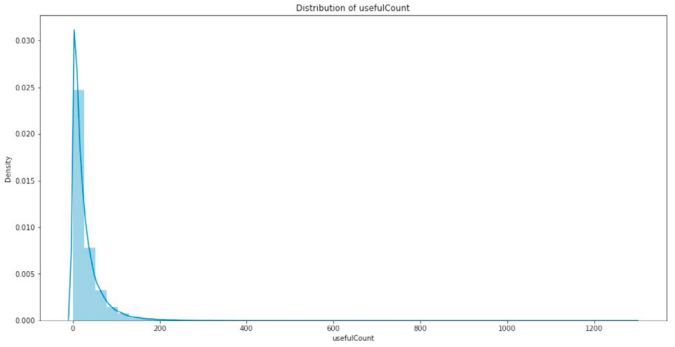


Fig. 3. Distribution of useful count

The fidelity of our dataset to this authoritative medical reference is paramount in enhancing the model's discernment of nuanced patterns and relationships within the data. By incorporating such reliable and comprehensive information, we aim to elevate the model's ability to capture relevant features essential for accurate predictions in drug recommendation. The substrate choice aligns seamlessly with our overarching goal of developing a sophisticated model that excels in providing precise and intelligent drug recommendations. This strategic alignment ensures that our dataset is well-equipped to meet the challenges of real-world healthcare scenarios.

Firstly, we preprocess the data. Preprocessing steps include removing any special symbols or characters from the text file, removing any stop words in the text, converting all the words into lowercase, lemmatizing all the wor.Fig 3 illustrates the distribution of review usefulness ('usefulCount') with the X-axis representing the count and the Y-axis denoting the density. Most reviews have a 'usefulCount' below 200, and as 'usefulCount' increases, the density declines, indicating fewer highly useful reviews.

VADER [24] emerges as a pivotal tool for sentiment analysis. Its significance lies in its adept combination of qualitative and quantitative methods, facilitating the construction and empirical validation of a platinum-standard list of lexical features. A lexicon, which refers to a stock of terms within a specific subject or language, plays a crucial role in this context. VADER's robust capabilities in sentiment analysis stem from its nuanced approach, making it a valuable asset in discerning sentiment in textual data. By establishing a thoughtfully tuned threshold, we effectively categorize these scores into positive and negative sentiments. This binary classification facilitates a nuanced understanding of user sentiments towards specific medications.

LIWC has about 4,500 words that are divided into 76 categories, including 905 terms specifically linked to sentiment analysis in two areas. In this method we use lexicon approach of Vader to find the sentiment of the various individuals who uses the specific medicines and categorize them in various bags of reliable data for example useful counts the Fig 4 shows the distribution of various words.

Our second step includes converting all the text into numerical sequences and also adding a padding. Text to sequence process is done using the 'text-to-sequence' method contained in the 'Tokenizer' class in Keras. We add padding by using 'pad-sequence' function in Keras. Padding is done to make all the texts are of the same length. This step is necessary because we have to define the input shape in deep learning models. The given below figure shows the most frquently used words which are tokenized to support a numerical format

3.2. Model Generation

The next step is dataset splitting, resulting in the categorization of the collected data into a training set and a test set. This is done to evaluate the performance of the model. As the name suggests, these data are utilized to both train and test the model.

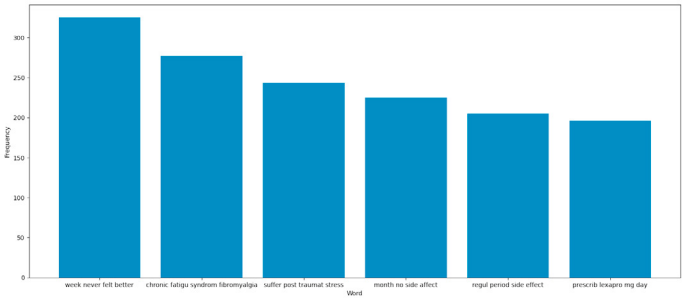


Fig. 4. Most Frequent Words

3.2.1. Random Forest

In training the Random Forest algorithm, we employ L. Breiman's original forest algorithm, where M trees are grown from random subsets of the original dataset. At each tree cell, the optimal split is determined by maximizing the CART criterion over a randomly chosen subset of directions (mtry). Predictions for a query point x involve averaging the target values of points within the same cell as x, as computed by each tree. The ultimate prediction relies on the pre-selected data points utilized in the tree-growing process.

3.2.2. Support Vector Machine

Support Vector Machines (SVM) leverage a majority of their features to create parallel partitions, employing two lines to delineate a distinctive space within a high-dimensional environment. At the core of SVM is the hyperplane, which generates hyperplanes with the utmost margin in the air, thereby establishing borders and effectively classifying data in this intricate high-dimensional space . The margin, representing the maximal separation between the nearest data points for two classes, plays a crucial role in SVM's classification prowess. As this margin widens, the generalization error of the classifier diminishes, underscoring the significance of this parameter in enhancing SVM's performance. Notably, among classifiers, SVM is recognized for its exceptional versatility, making it a valuable tool in various applications [11].

3.2.3. Proposed LSTM

The proposed LSTM model introduces distinctive features, building upon the standard setup. It initiates with an embedding layer, skillfully converting integer-encoded vocabulary indices into fixed-dimensional vectors. Following this, two LSTM layers are thoughtfully incorporated, each with 128 units for seamless sequential data processing. To gracefully handle the training phase, dropout layers are judiciously placed after each LSTM layer. A notable improvement is observed in the first LSTM layer,It permits a more thorough interpretation of the material by retaining all sequence information. The model then gracefully concludes with dense layers, featuring 100 units and ReLU activation, followed by a singular sigmoid-activated unit. This thoughtful refinement positions the LSTM model as an evolved solution, leveraging specific choices in hyperparameters and layer configurations, promising enhanced performance in binary sentiment classification within the landscape of sequential data processing.

our methodology acknowledges the dynamic nature of healthcare scenarios, prompting a commitment to regular model updates. Recognizing that medical information evolves, routine updates are essential to ensure that the drug recommendation system stays abreast of the latest developments in pharmacology and healthcare practices. This iterative approach guarantees the continued relevance and effectiveness of our system

3.3. Testing and Evaluation

Final step is evaluation of the model. Here we use accuracy score, confusion matrix, precision, recall and F1 score to evaluate the performance of the model. We determine the above matrices with the help of our testing set.we adopt several strategic measures to bolster model performance. To fine-tune the model's behavior, we delve into the intricate process of adjusting classification thresholds, allowing us to optimize precision or recall based on specific application needs. We also evaluate each of the three models with these matrices.

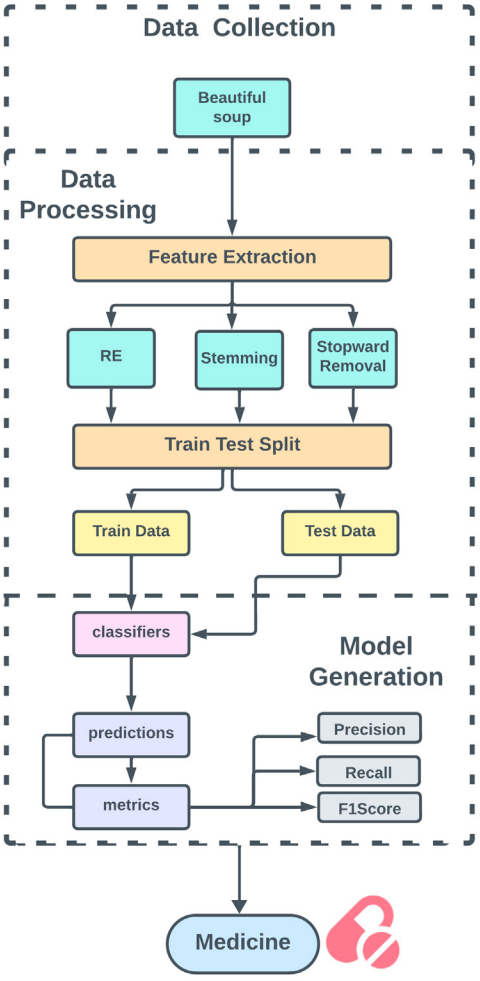


Fig. 5. Detailed flow of the proposed model

4. Result Analysis

Result analysis in this experiment involves a thorough examination of the outcomes produced by the drug recommendation system's models, including Random Forest, SVM, and the proposed LSTM network. We will evaluate the models' accuracy, precision, recall, and F1 score during this crucial stage, which gives us a thorough grasp of how well they work. Analysis of the results is essential since it not only measures the models' efficacy but also clarifies their advantages and disadvantages. While precision emphasises the accuracy of positive predictions, accuracy score shows the percentage of correctly identified labels. Recall evaluates the model.

4.1. Performance Comparison

In our conducted experiment, we employed the specified tools and implemented the Random Forest method on our dataset, achieving 20 accurately classified drugs out of 20 with an impressive 87% accuracy within a rapid 120 milliseconds. Subsequently, we applied the Support Vector Machine classifier algorithm, yielding 20 correct classifications out of 20, and attaining an accuracy of 85% in a swift 2.99 milliseconds. Furthermore, we explored the

Table 1. Model Performance

Model	Class	Precision	Recall	F1 Score
RFC	Negative	0.86	0.77	0.81
	Positive	0.91	0.95	0.93
SVM	Negative	0.84	0.87	0.86
	Positive	0.81	0.93	0.94
Proposed Model	Negative	0.89	0.84	0.86
	Positive	0.94	0.706	0.75

application of a deep learning model featuring LSTM layers on our dataset, resulting in the correct classification of all 20 drugs with a notable accuracy of 94% within a slightly longer timeframe of 10.4 seconds.

4.2. Confusion Matrix

A confusion matrix is used to visually represent the actual and predicted values. It visualizes the performance of our classification models and it is represented in the Table 1. Here True Positive is termed TP, True Negative is TN, False positive is FP, and False Negative is FN. True positives are instances correctly identified as positive, true negatives are instances correctly identified as negative, and false positives are instances incorrectly identified as positive [8].

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F1\ Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

The proposed model excels in precision for positive predictions, achieving 0.94 compared to RFC's 0.91 and SVM's 0.81. While its recall for the positive class is slightly lower at 0.706 versus RFC's 0.95, the model maintains a balanced F1 Score of 0.75, showcasing effective positive predictions with a low false positive rate. Overall, the model underscores superior precision for positive instances, emphasizing its accuracy. Task-specific considerations are crucial for model selection.

The proposed model enhances real-time drug recommendations by leveraging its superior precision in identifying positive instances, ensuring more accurate predictions based on patient symptoms. This capability contributes to a more effective and tailored drug recommendation system, optimizing patient outcomes in real-time scenarios.

Conclusion

In this research, we present an interactive framework which promotes the fitting to find a safe and suitable drug/medication to the patient according to the symptoms or indications exhibited by the patient. Drugs dataset has been manually created from CIMS drug monitor drug reference book. Moreover, we have conducted various experiments using different data mining methods. All three algorithms that we used in this experiment, which are Random Forest Classifier, Support Vector Machine and modified-LSTM, Propsed LSTM model have shown 94 percent in terms of accuracy. It is essential to emphasize that the aim of this paper is to illustrate the methodology for extracting sentiment from data and building a recommendation system through classification. Additionally, hyperparameter

optimization is necessary to improve the accuracy of the classification algorithms.Feature engineering is pivotal in model development, focusing on crafting and selecting pertinent features for enhanced performance. This aimed to capture nuanced patterns, refining data relationships for heightened predictive accuracy.Further improvements and optimizations are necessary for the system to perform effectively in real-life scenarios.

References

[1] H. Sak, A. Senior, and F. Beaufays, "Long short-term memory recurrent neural network architectures for large scale acoustic modeling," in *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*, 2014, pp. 338–342.

[2] X. Chen, P. Cong, and S. Lv, "A long-text classification method of chinese news based on bert and cnn," *IEEE Access*, vol. 10, pp. 34 046–34 057, 2022.

[3] S. Garg, "Drug recommendation system based on sentiment analysis of drug reviews using machine learning," in *11th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, 2021, pp. 175–181.

[4] G. W. Milligan and S. C. Hirtle, "Clustering and classification methods," *Handbook of Psychology: Research Methods in Psychology*, vol. 2, pp. 165–186, 2003.

[5] S. Ghosh, A. Dasgupta, and A. Swetapadma, "A study on support vector machine based linear and non-linear pattern classification," in *2019 International Conference on Intelligent Sustainable Systems (ICISS)*, 2019, pp. 24–28.

[6] J. L. Speiser, M. E. Miller, J. Tooze, and E. Ip, "A comparison of random forest variable selection methods for classification prediction modeling," *Expert Systems with Applications*, vol. 134, pp. 93–101, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417419303574

[7] L. Rahul, L. S. Meetei, and H. S. Jayanna, "Statistical and neural machine translation for manipuri-english on intelligence domain," *Lecture Notes in Electrical Engineering*, 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:236715478

[8] S. Visa, B. Ramsay, A. Ralescu, and E. Knaap, "Confusion matrix-based feature selection." vol. 710, 01 2011, pp. 120–127.

[9] J. D. Novaković, A. Veljović, S. S. Ilić, Papić, and M. Tomović, "Evaluation of classification models in machine learning," 2017, (Spring 2017): 39-46.

[10] N. M. Lal, S. Krishnanunni, V. Vijayakumar, N. Vaishnavi, S. Siji Rani, and K. Deepa Raj, "A novel approach to text summarisation using topic modelling and noun phrase extraction," in *Advances in Computing and Network Communications: Proceedings of CoCoNet 2020, Volume 2*. Springer, 2021, pp. 285–298.

[11] Zhixiang, Z. Xu, J. Gardner, S. Tyree, and K. Weinberger, "Compressed support vector machines," 01 2015.

[12] M. Pyingkodi, K. Thenmozhi, K. Chitra, M. Karthikeyan, M. Pradeep, P. T. D. Suriya, and T. J. Viswanathan, "Hate speech analysis using supervised machine learning techniques," in *2023 International Conference on Computer Communication and Informatics (ICCCI)*, 2023, pp. 1–6.

[13] C. Hutto and E. Gilbert, "Vader: A parsimonious rule-based model for sentiment analysis of social media text," in *Proceedings of the 8th International Conference on Weblogs and Social Media, ICWSM 2014*, 2015.

[14] Y. K. M, S. C K, T. A R, S. B. Kumar, and G. Sarath, "A twitter-based software vulnerability alert framework using natural language processing," in *2023 8th International Conference on Communication and Electronics Systems (ICCES)*, 2023, pp. 702–707.

[15] S. B. Kotsiantis, "Supervised machine learning: A review of classification techniques," June 2007.

[16] S. S. Baby and S. Siji Rani, "A hybrid product recommendation system based on weather analysis," in *Proceedings of the 6th International Conference on Advance Computing and Intelligent Engineering*, B. Pati, C. R. Panigrahi, P. Mohapatra, and K.-C. Li, Eds. Singapore: Springer Nature Singapore, 2023, pp. 263–273.

[17] A. Singh, S. K. Dargar, A. Gupta, A. Kumar, A. K. Srivastava, M. Srivastava, P. K. Tiwari, and M. A. Ullah, "Evolving long short-term memory network-based text classification," 2022.

[18] S. Siji Rani, P. Dhanyalaxmi, A. Akshay, K. Ananthakrishnan, and A. Siva Sankar, "Spatio-temporal prediction in epidemiology using graph convolution network," in *International Conference on Information and Communication Technology for Intelligent Systems*. Springer, 2023, pp. 367–378.

[19] K. Smagulova and A. James, "A survey on lstm memristive neural network architectures and applications," *European Physical Journal Special Topics*, vol. 228, pp. 2313–2324, 2019.

[20] T. T. Ajith, A. K. CV, J. Nandakishore, M. A. Subramanian *et al.*, "Enhanced movie recommendation using knowledge graph and particle filtering," in *2021 2nd International Conference on Smart Electronics and Communication (ICOSEC)*. IEEE, 2021, pp. 1139–1144.

[21] T. Davenport and R. Kalakota, "The potential for artificial intelligence in healthcare," *Future Healthcare Journal*, vol. 6, no. 2, pp. 94–98, Jun 2019. [Online]. Available: https://doi.org/10.7861/futurehosp.6-2-94

[22] S. M. D. Thumati, K. Dhanya, H. Sathish, K. S. Madan, and S. Rani, "A comparative study on the working of gnn and cnn on panoramic x-rays in prediction of dental diseases," in *2023 8th International Conference on Communication and Electronics Systems (ICCES)*. IEEE, 2023, pp. 755–762.

[23] S. Minaee, N. Kalchbrenner, E. Cambria, N. Nikzad, M. Chenaghlu, and J. Gao, "Deep learning based text classification: A comprehensive review," 2020, 6 Apr 2020 (v1).

[24] A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola, "Dive into deep learning," *CoRR*, vol. abs/2106.11342, 2021. [Online]. Available: https://arxiv.org/abs/2106.11342